INCH-POUND

MIL-PRF-19500/730
6 September 2005

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, SCHOTTKY, DUAL, CENTER TAP,
TYPES 1N7037CCU1, 1N7043CAT1, 1N7043CCT1, JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, schottky power rectifier diodes. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (U1) and figure 2 (TO-254AA).

1.3 Maximum ratings. Unless otherwise specified, maximum ratings (T_C = +25°C).

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6
Types (1)	V_{RWM}	I_O (2)(3) T_C = +100°C	I_{FSM} (3) t_p = 8.3 ms, T_C = +25°C	$R_{\theta JC}$ (4)	T_{STG} and T_J
	V dc	A dc	A (pk)	°C/W	°C
1N7037CCU1	100	35	250	0.8	-65 to +150
1N7043CCT1 1N7043CAT1	100	35	175	1.15	

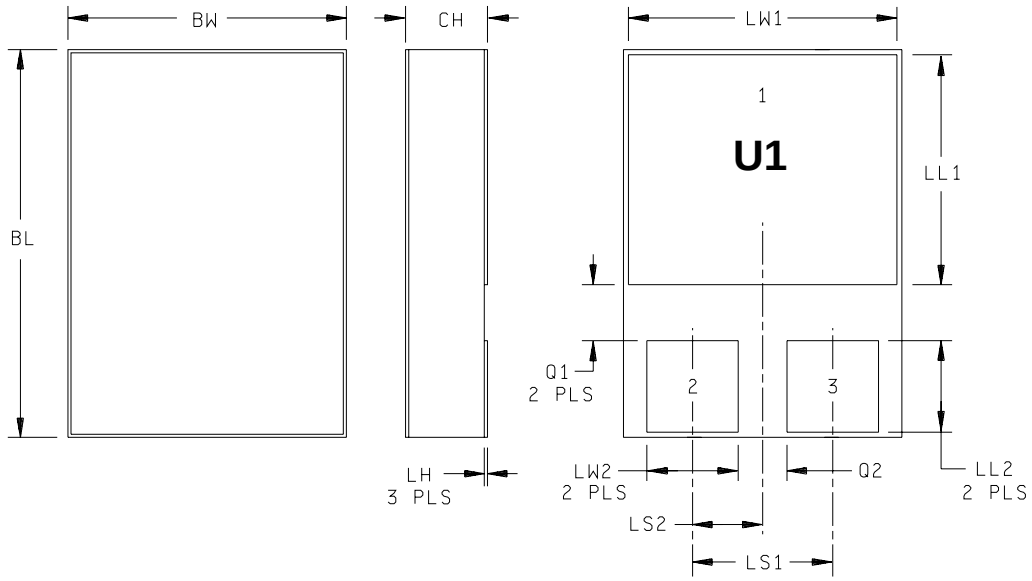
(1) Electrical characteristics are the same for all package styles and polarity configurations.
(2) See temperature-current derating curves in figures 3 and 5.
(3) Per leg.
(4) Per package.

1.4 Primary electrical characteristics. $R_{\Theta JC}$ = 1.6°C/W maximum for 1N7037CCU1, $R_{\Theta JC}$ = 2.3°C/W maximum for 1N7043CCT1 and 1N7043CAT1, $R_{\Theta JA}$ = 40°C/W maximum, each individual diode.

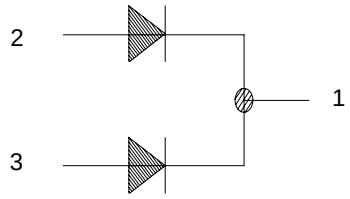
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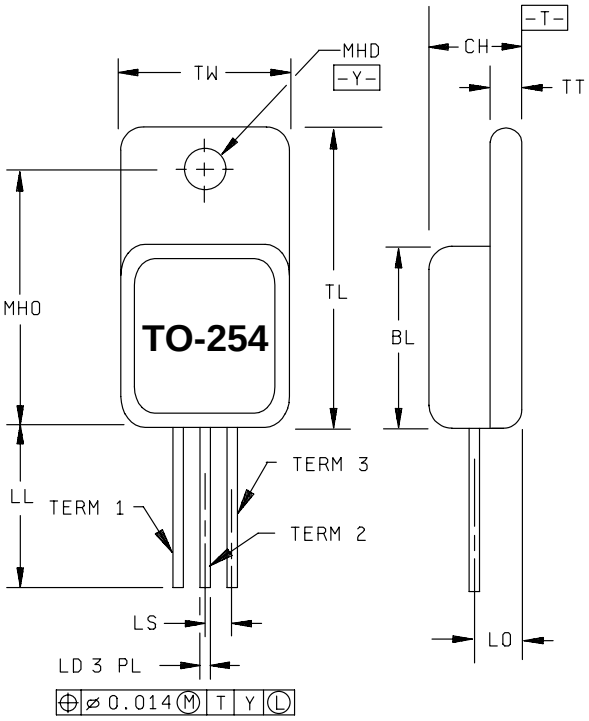
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.620	.630	15.75	16.00
BW	.445	.455	11.30	11.56
CH	.129	.139	3.28	3.53
LH	.010	.020	0.26	0.51
LW1	.370	.380	9.40	9.65
LW2	.135	.145	3.43	3.68
LL1	.410	.420	10.41	10.67
LL2	.152	.162	3.86	4.12
LS1	.200	.220	5.08	5.59
LS2	.100	.110	2.54	2.79
Q1	.035		0.89	
Q2	.050		1.27	



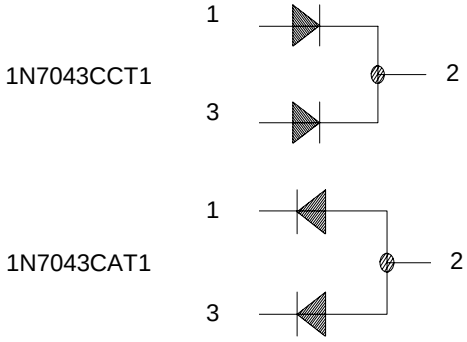
NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 1. Dimensions and configuration 1N7037CCU1.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.84
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.510	.570	12.95	14.48
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.81 BSC	
MHD	.139	.149	3.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.84



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. In accordance with ASME Y14.5M, diameters are equivalent to Φx symbology.

FIGURE 2. Dimensions and configuration 1N7043CAT1 and 1N7043CCT1 (TO-254AA).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 2 herein. Methods used for electrical isolation of the terminal feedthroughs for the TO-254 shall employ materials that contain a minimum of 90 percent Al_2O_3 (ceramic).

3.4.1 Polarity. Polarity and terminal configuration shall be in accordance with figures 1 and 2 herein.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.6 Electrical test requirements. The electrical test requirements shall be as specified in tables I and II herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500 and herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II herein).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

4.3 Screening (JANS, JANTXV and JANTX levels). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3b	Condition A, one pulse, I_O = 0, V_{RWM} = 0, See 1.3 Column 4.	Condition A, one pulse, I_O = 0, V_{RWM} = 0, See 1.3 Column 4.
(1) 3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
3d	Peak reverse energy test (see 4.3.3)	Peak reverse energy test (see 4.3.3)
4	Required	Optional
5	Required	Not applicable
7	Optional	Optional
8	Required	Not applicable
9, 10	Not applicable	Not applicable
11	V_{F1} and I_{R1}	V_{F1} and I_{R1}
12	see 4.3.1	see 4.3.1
13	Subgroup 2 and 3, of table I herein, V_{F1} and I_{R1}, excluding thermal impedance; ΔV_{F1} = ±50 mV (pk); ΔI_{R1} = ± 100 percent from the initial value.	Subgroup 2, of table I herein excluding thermal impedance; V_{F1} and I_{R1}; ΔV_{F1} = ±50 mV (pk); ΔI_{R1} = ± 100 percent from the initial value.
14	Required	Required
15	Required	Not applicable
16	Required	Not applicable

(1) Thermal impedance shall be performed any time after screen 3.

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4.3.1 Power burn-in conditions. Burn-in conditions are as follows: Method 1038 of MIL-STD-750, test condition A. T_C = +125°C; V_R = 80 V dc.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). The thermal impedance limit used in 4.3, screen 3c and the subgroup 2 of table I shall comply with the thermal impedance graphs on figures 4 and 6 (less than or equal to the curve value at the same t_H time) and shall be less than the process determined statistical maximum limit as outlined in method 3101. See group E, subgroup 4 herein.

4.3.3 Peak reverse energy test. The peak reverse energy test is to be performed using the circuit as shown on figure 7 or equivalent. The Schottky rectifier under test must be capable of absorbing the reverse energy, as follows: I_{RM} = 1A minimum, V_{RSM} = 100V minimum, L = 100μH.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500, and table I herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) and VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, forward voltage test (V_{F1}) and reverse leakage test (I_{R1}) herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	ΔT_C = +85°C, I_F = 2 A minimum for 2,000 cycles.
B5	1038	Condition B, I_F = 1 A dc minimum, adjust T_A and I_F to achieve T_J = +150°C, +0° -10°C, t = 240 hours minimum; (heat sinking allowed).
B6	3101	Limit for thermal resistance for 1N7037 is 1.6°C/W per side. Limit for thermal resistance for 1N7043 is 2.3°C/W per side.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1037	ΔT_C = +85°C minimum, I_F = 2 A minimum for 2,000 cycles.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2, forward voltage test (V_{F1}) and reverse leakage test (I_{R1}) herein. Delta measurements shall be in accordance with table II herein.

Subgroup	Method	Condition
C2	2036	Condition A, weight = 10 lbs, t = 15 seconds. Not applicable to U1 package.
C6	1037	ΔT_C = +85°C, minimum, I_F = 2 A minimum for 6,000 cycles.

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4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table IX of MIL-PRF-19500, and table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 1N7037 1N7043	3101	See 4.3.2	$Z_{\Theta JX}$		1.45 2.15	˚C/W
Forward voltage	4011	Pulsed test (see 4.5.1)	V_{F1}			
1N7037 1N7043		I_F = 15A (pk) I_F = 15A (pk)			.90 .95	V V
Forward voltage	4011	Pulsed test (see 4.5.1)	V_{F2}			
1N7037 1N7043		I_F = 35 A(pk) I_F = 35 A(pk)			1.22 1.30	V V
Reverse current	4016	DC method	I_{R1}			
1N7037 1N7043		V_R = 100 V V_R = 100 V			.5 .5	mA mA
Subgroup 3						
High temperature operation:		T_C = +125 °C				
Forward voltage	4011	Pulsed test (see 4.5.1)	V_{F3}			
1N7037 1N7043		I_F = 35 A(pk) I_F = 35 A(pk)			1.00 1.20	V V
Reverse current	4016	DC method;	I_{R2}			
1N7037 1N7043		V_R = 100 V V_R = 100 V			15 15	mA mA
Low temperature operation:		T_C = -55°C				
Forward voltage	4011	Pulsed test (see 4.5.1)	V_{F3}			
1N7037 1N7043		I_F = 35 A dc I_F = 35 A dc			1.35 1.84	V V

See footnotes at end of table.

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TABLE I. Group A inspection – Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Junction capacitance	4001	V_R = 5 V dc, f = 1 MHz, V_{SIG} = 50 mV (p-p)	C_J		600	pF
Subgroup 5						
Not applicable						
Subgroup 6						
Surge	4066	See column 4 of 1.3. 10 surges per diode. 60 seconds between surges. (see 4.5.1)				
Electrical measurements		See table I, subgroup 2 herein.				
Subgroup 7						
Dielectric withstanding voltage	1016	V_R = 500 V dc; All leads shorted; V measure from leads to case;	DWV		10	µA
Scope display evaluation	4023	Stable only				
Electrical measurements		See table I, subgroup 2 herein.				

1/ For sampling plan, see MIL-PRF-19500.
2/ Electrical characteristics apply to all package styles and polarities.

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TABLE II. Groups B, C, and E delta requirements. 1/ 2/ 3/ 4/ 5/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_F = 15 A (pk) pulsed (see 4.5.1)	ΔV_{F1}	± 50 mV dc from initial reading.		
2.	Reverse current	4016	Vr = 100V	ΔI_{R1}	±100% from initial reading.		
3.	Thermal impedance	3101	See 4.3.2	$Z_{\Theta JX}$			°C/W

1/ Each individual diode.
2/ The electrical measurements for appendix E, table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 4, see table II herein, steps 1, 2, and 3.
   b. Subgroup 5, see table II herein, steps 1 and 2.
3/ The electrical measurements for appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table II herein, steps 1 and 2.
   b. Subgroup 3, see table II herein, steps 1, 2, and 3.
   c. Subgroup 6, see table II herein, steps 1 and 2.
4/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:
   a. Subgroups 2 and 3, see table II herein, steps 1 and 2 for all levels.
   b. Subgroup 6, see table II herein, steps 1, 2, and 3 for all levels.
5/ Devices which exceed the table I, limits for this test shall not be accepted.

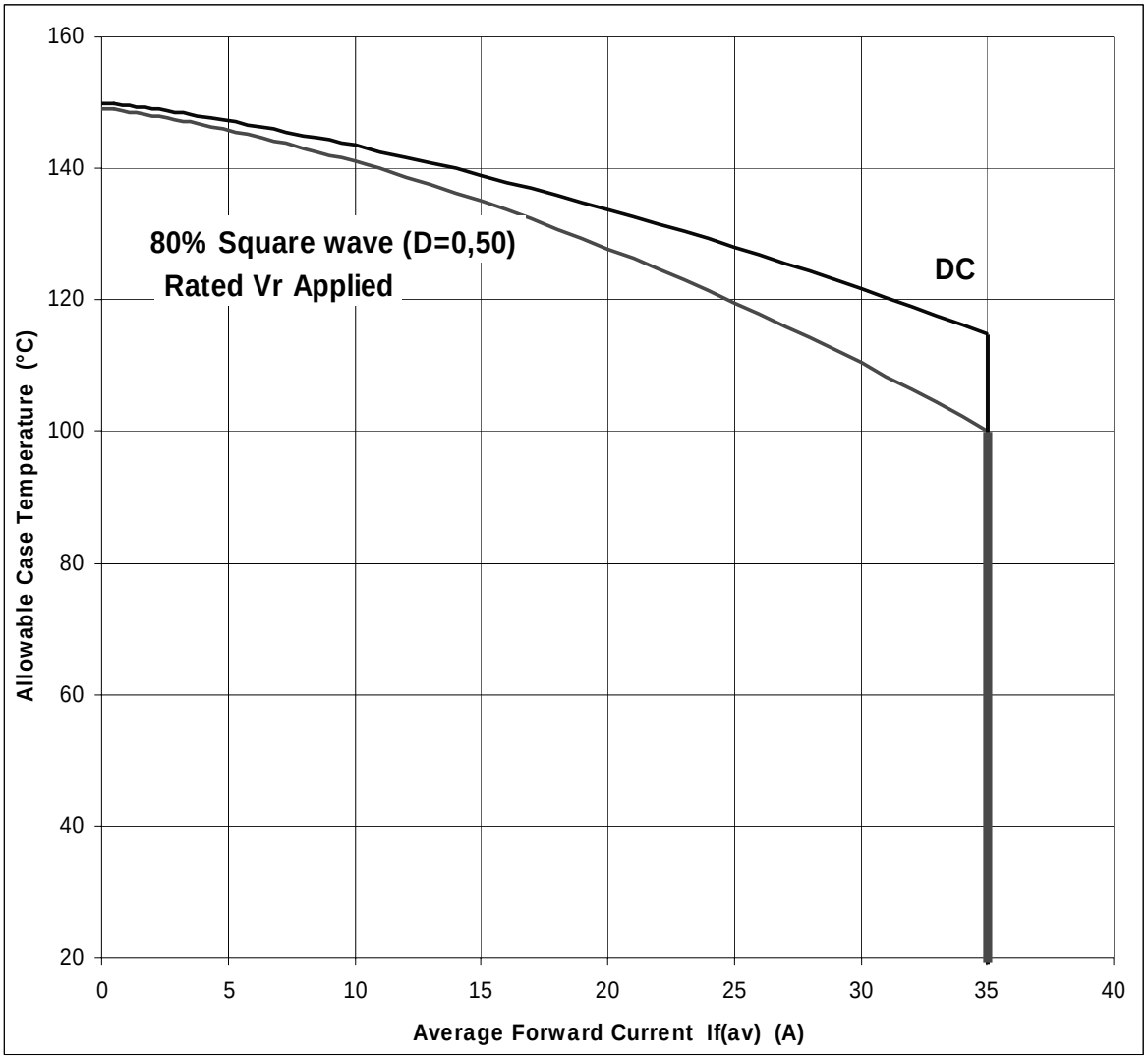
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TABLE III. Group E inspection (all quality levels) – for qualification and requalification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			n = 12, c = 0
Temperature cycling (air to air)	1051	Test condition G, 500 cycles, -55°C to +150°C.	
Electrical measurements		See table I subgroup 2.	
Subgroup 2			n = 12, c = 0
Life test	1048	t = 1,000 hours, T_C = +125°C, V_R = 80 percent rated voltage (see 1.3, column 2 herein).	
Electrical measurements		See table I subgroup 2.	
Subgroup 4			
Thermal impedance curves		Each supplier shall submit their qualification lot average and design-maximum thermal impedance curves to the qualifying activity. In addition, the optimal test conditions and thermal impedance limit shall be provided to the qualifying activity in the qualification report.	
Subgroup 6 1/			n = 5, c = 0
Surge	4066		
1N7037		Condition A, T_A = +25°C I_{FSM} = 250 A, 10 surge of 8.3 ms superimposed on I_O. V_R = 0; I_O = 10 A pk half sine wave, continuous.	
1N7043		Condition A, T_A = +25°C I_{FSM} = 175 A, 10 surge of 8.3 ms superimposed on I_O. V_R = 0; I_O = 10 A pk half sine wave, continuous.	
Electrical measurements		See table I subgroup 2 (Vf and Ir only).	
Subgroup 7			n = 3, c = 0
ESD	1020		

1/ Each individual diode.

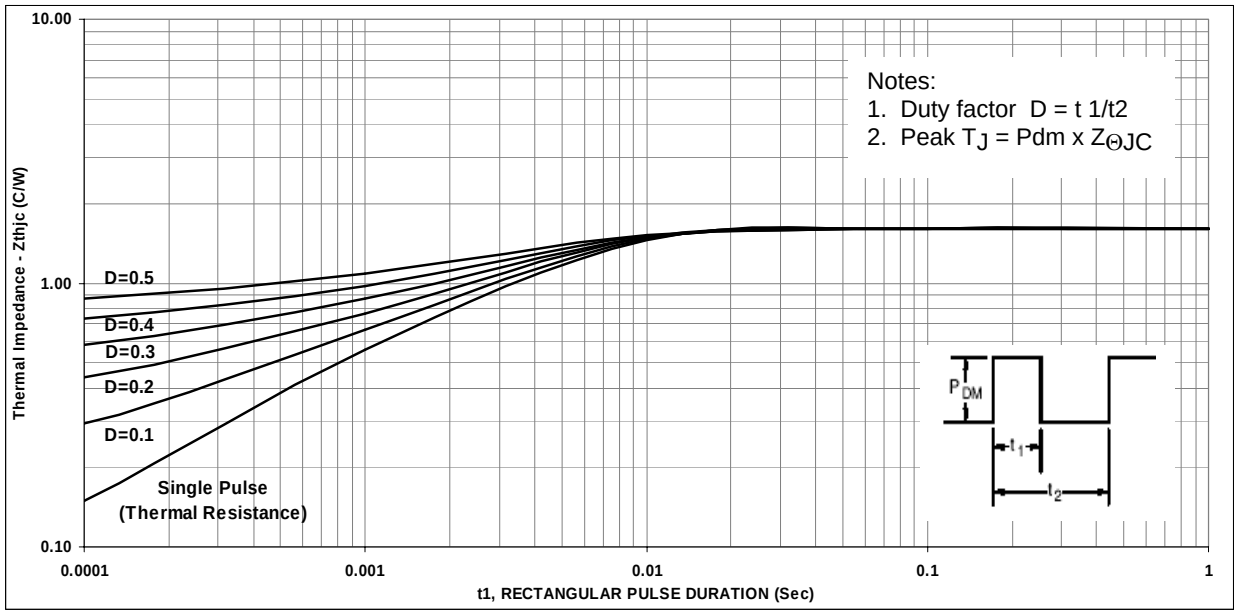
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Rating is per package.

Figure 3. Temperature-current derating curve 1N7037.

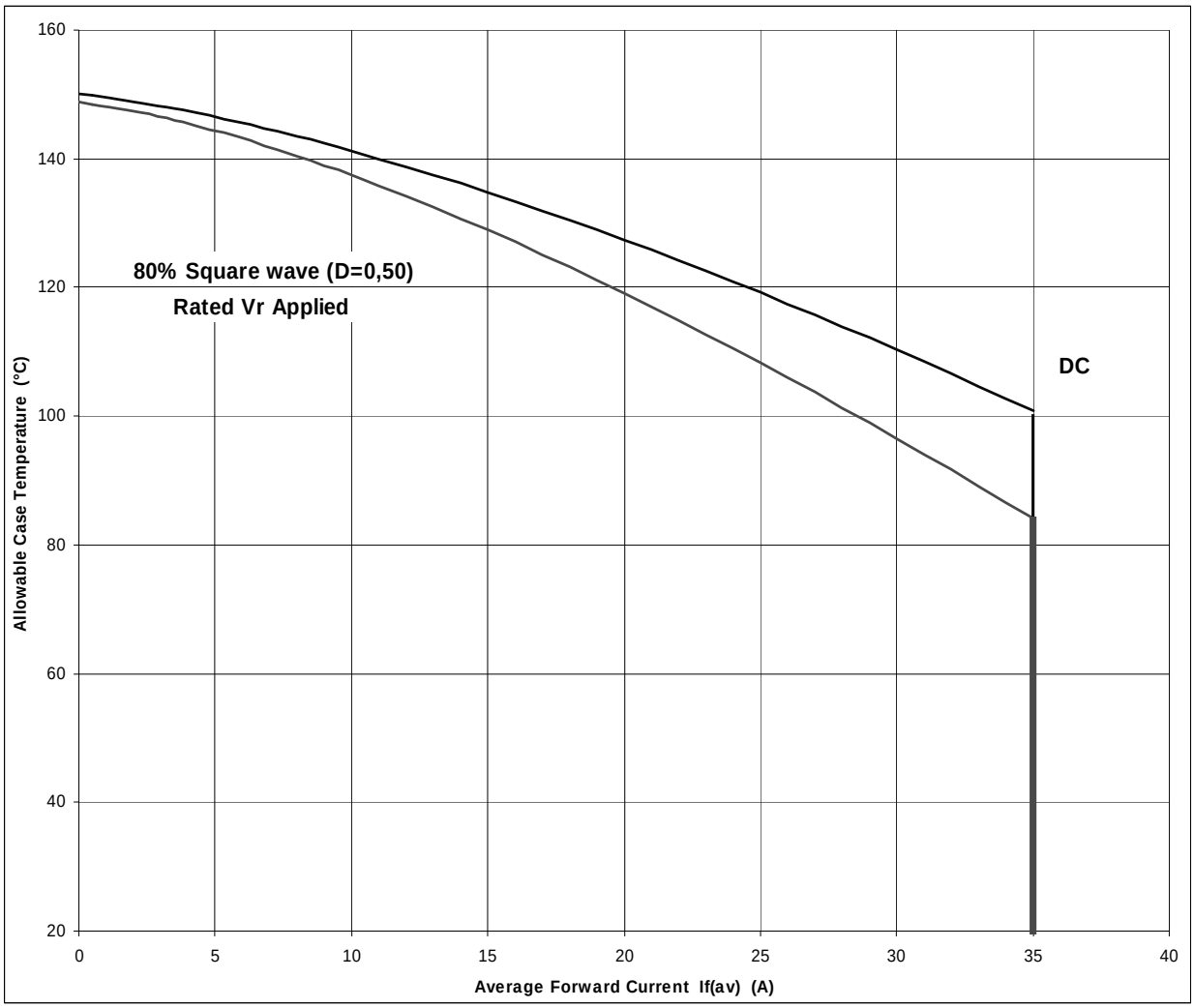
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Rating is per leg.

Figure 4. Thermal impedance 1N7037.

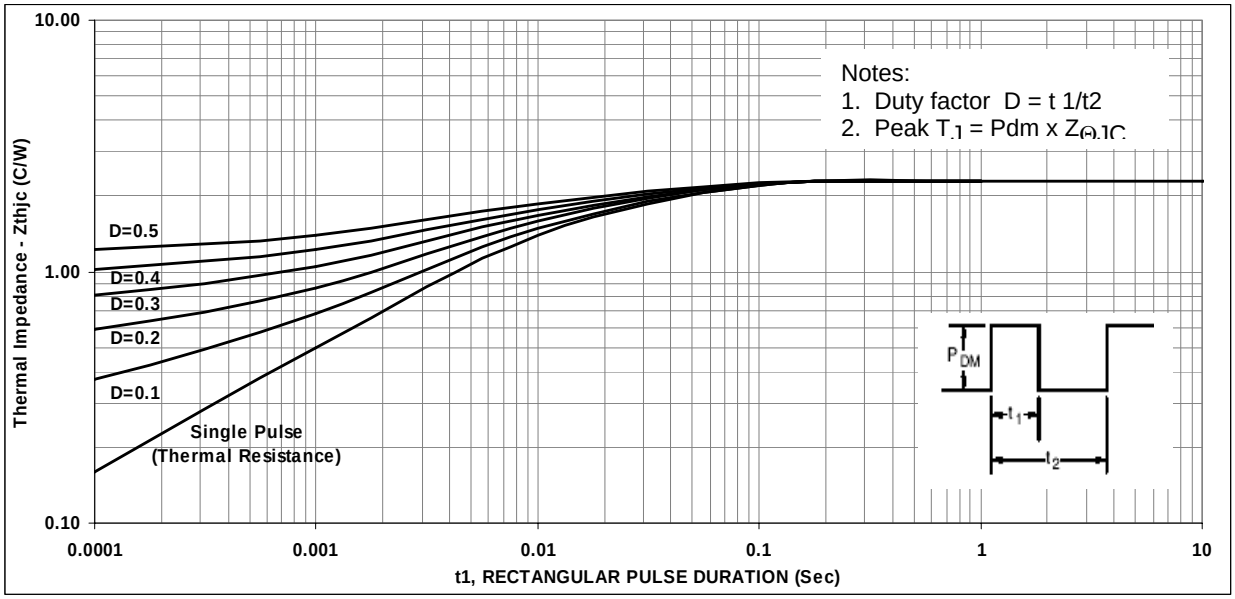
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Rating is per package.

Figure 5 Temperature-current derating curve 1N7043.

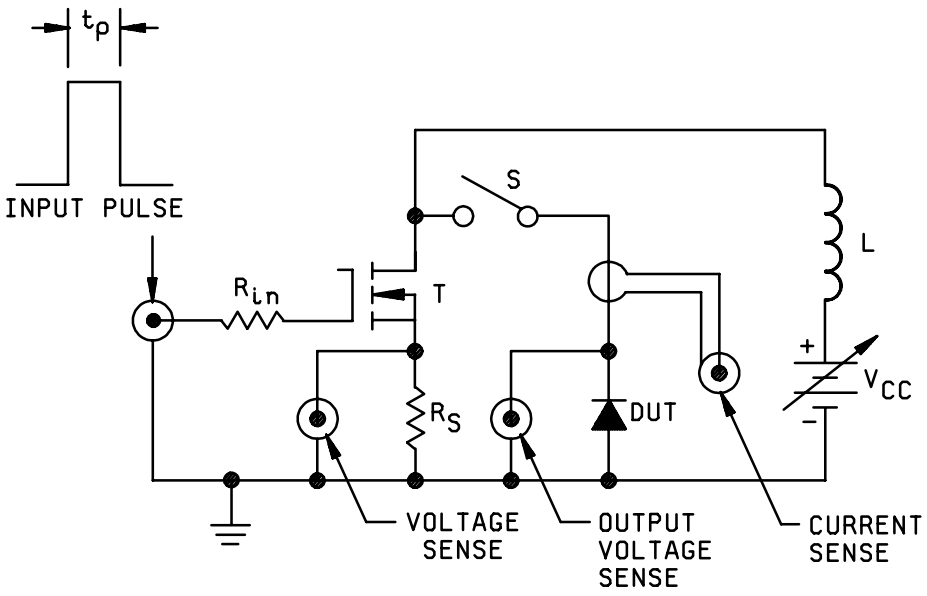
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Rating is per leg.

Figure 6. Thermal impedance 1N7043.

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Input pulse R_{in} = 50 ohms, 1 watt
V_G = 10 Volts, R_S = 0.1 ohms, 1 watt
Z_G = 50 ohms
L = 100μH
P.W. ≈ 30 μs
Duty cycle ≤ 1 percent, T = IRF250/2N6766 or equivalent

PROCEDURES:

1. With S open, adjust pulse width to test current of 1 amps across R_S.
2. Close S, verify test current with current sense.
3. Read peak output voltage (see 4.3.3).

FIGURE 7. Peak reverse energy test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Cross reference substitution list. A PIN for PIN replacement table follows, and these devices are directly interchangeable.

Non-preferred PIN	Preferred PIN
15CGQ100	JANS, JANTXV, JANTX1N7043CCT1
15JGQ100	JANS, JANTXV, JANTX1N7043CAT1
15CLQ100	JANS, JANTXV, JANTX1N7037CCU1

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2927)

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